

Network Architecture

Lenuta Alboaie
lenuta.alboaie@info.uaic.ro

Content

- Structure of Computer Networks
- Network Architecture Models (OSI, TCP/IP)
- TCP/IP Model
- ISO/OSI versus TCP/IP

Computers Networks Structure

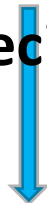
- Computer Network Structure – **stack levels**

- Functionality :

- **Interface**: ensure communication between two consecutive levels
- **Service**: functionality provided by a level

Result: reducing design complexity

- The principle of communication: **transmitter sends at n level what the recipient receives at the n level**



- **Protocol** – Rules and conventions through which the communication takes place

Example: link among - levels, protocols and interfaces

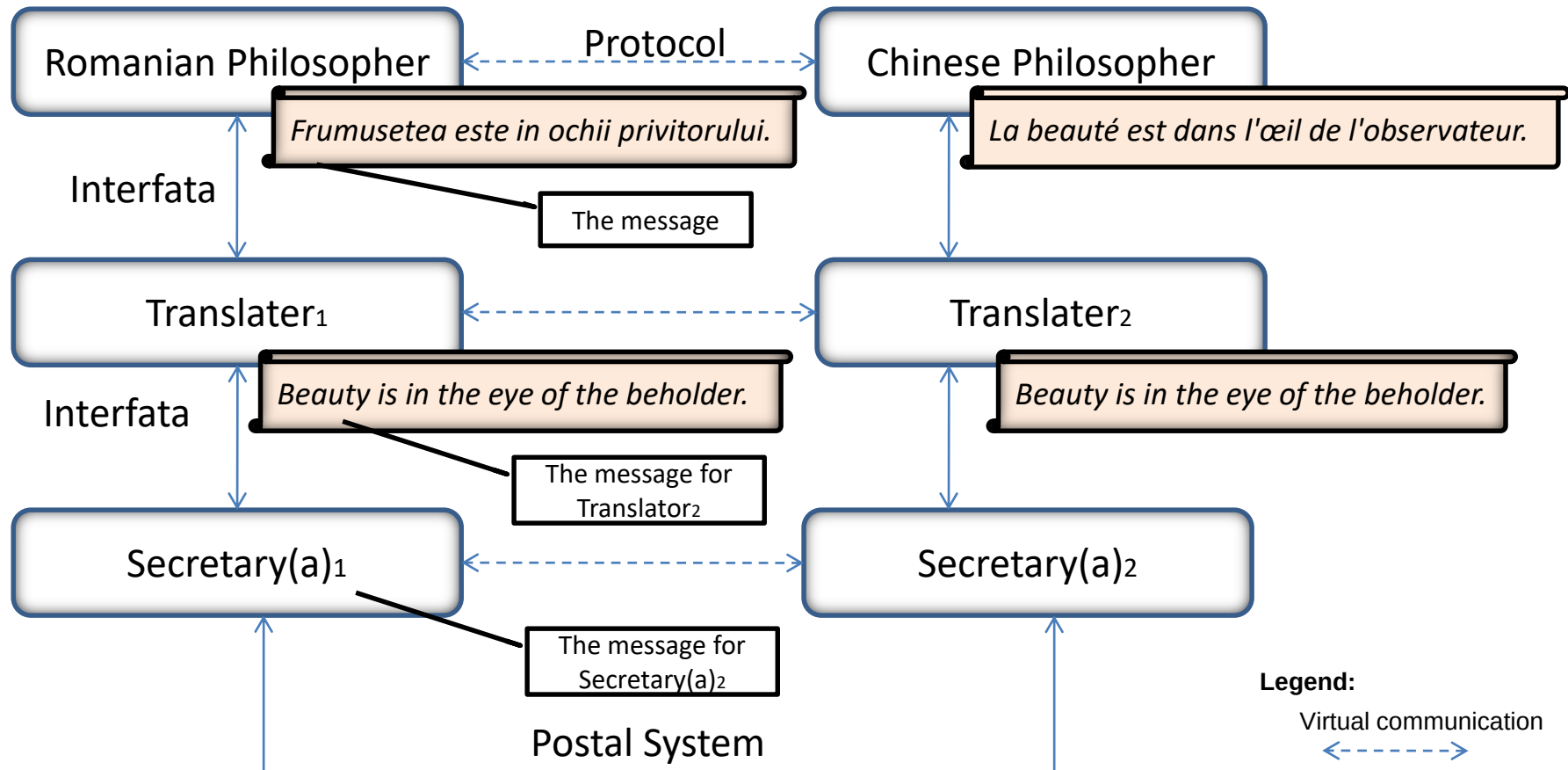
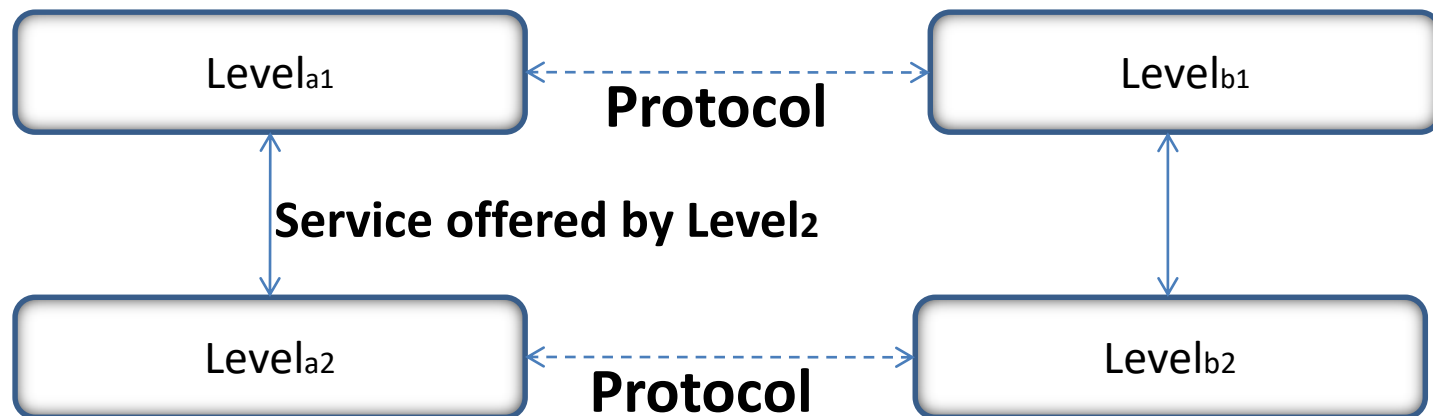


Figure: Architecture philosophe – translator - secretary

Aspects regarding levels design

- Specifying the service is performed by a set of primitives (operations) available to the one who uses the service
- **Service!= Protocol**



Aspects regarding levels designs

- Services types
 - *Connection-oriented*
 - Communication requires a connection
 - Similar to a telephone service
 - *Connectionless*
 - Communication does not require a connection
 - Similar to postal service

Aspects regarding level design

- **Network architecture:** the set of levels and protocols
 - Architecture specification must provide sufficient information for programs or equipment intended, in order to offer the specific protocols
- **Protocols stack:** list of protocols (on all levels) used by a particular system



Aspects regarding levels design

- Each level must identify transmitters & receivers through an *addressing mechanism*
- Data transfer rules identification
 - simplex communication
 - Example: TV
 - half-duplex communication
 - Example: "walkie-talkie"
 - full-duplex communication
 - Example: telephone

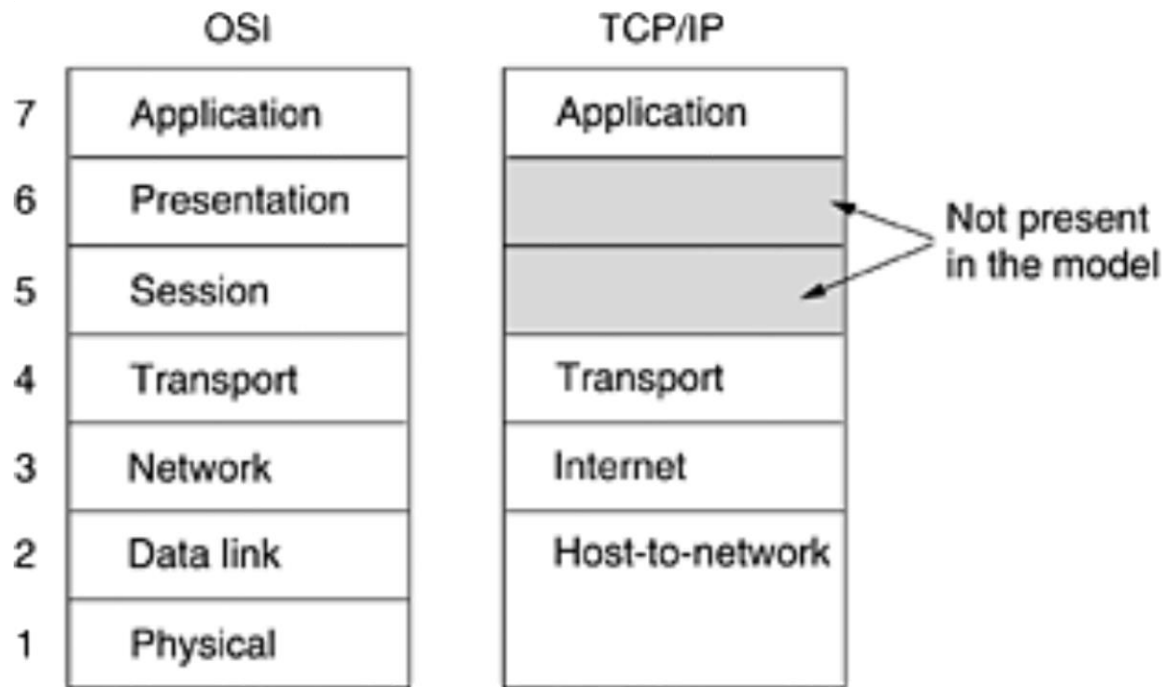


Aspects regarding levels design

- In general, communication channels do not preserve the order of the sent messages => need for a protocol that provides a mechanism to reconstruct the correct messages order
- Sometimes the receiver cannot manage the variable length messages => it must be a mechanism to divide/assemble
- High costs in the allocation of separate connections? => Multiplexing - use the same connection for independent conversations
- In general, there are several ways between source and destination => routing mechanism
- Physical communication circuits are not perfect => it asks for an error control mechanism

Reference models for network architecture

- **ISO/OSI** (*International Standard Organization/ Open System Interconnection*)
- **TCP/IP** (*Transmission Control Protocol/ Internet Protocol*)



[Computer Networks, 2010 – Andrew S. Tanenbaum, et.al.]

Network Architecture - Equipment

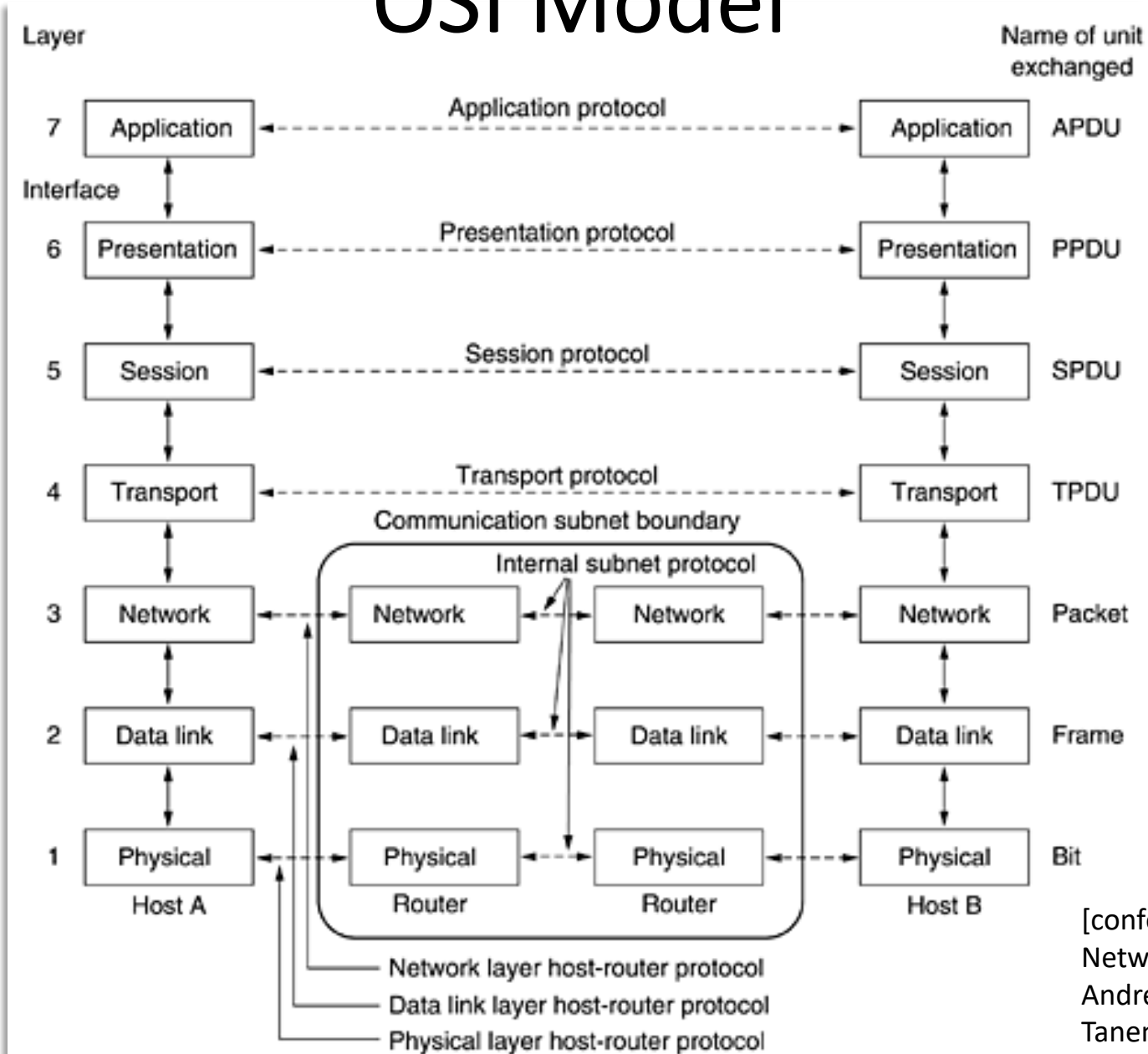
Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

Figure: Devices and appropriate levels

OSI Model- motivation

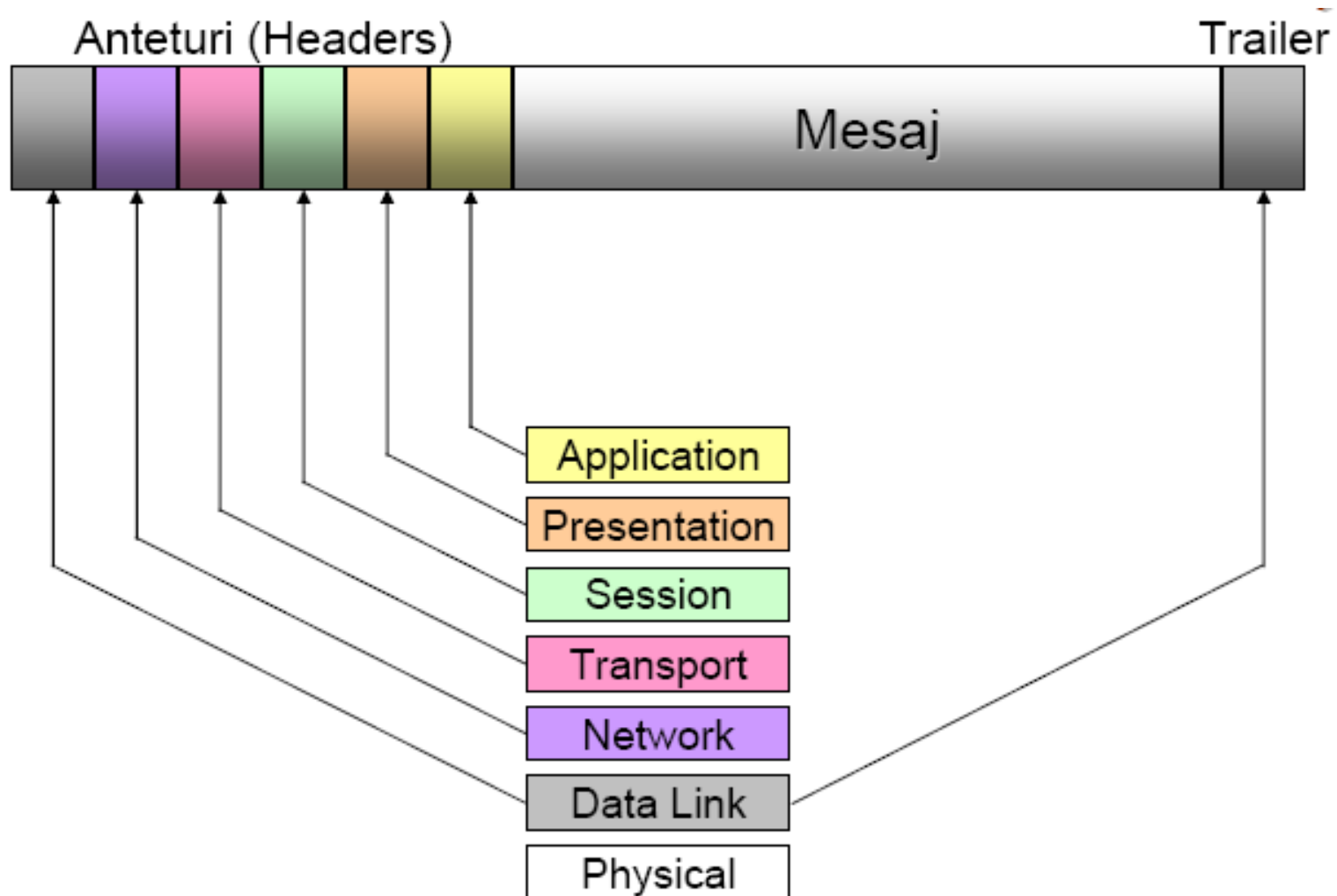
- The need for a different level of abstraction => to create a new level
 - Obs. The number of levels must be optimal, therefore each level has different functions and the whole architecture is functional
- A level has a clear role; a level function must take into account protocols that are standardized at international level
- Minimizing the flow of information between levels is accomplished through good boundary levels => Levels can be modified and implemented independently
- Each level offers services for superior level (using services from previous levels)
- “*peer*” levels of different systems communicate via a protocol

OSI Model



[conform Computer Networks, 2010 – Andrew S. Tanenbaum, et.al.]

OSI Model – message structure

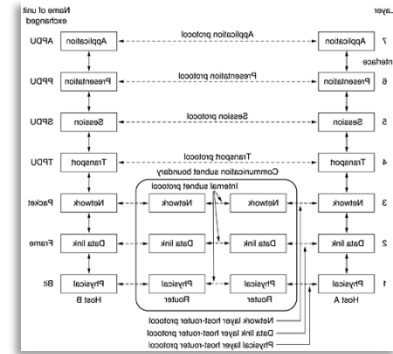


[Retele de calculatoare – curs 2007-2008, Sabin Buraga]

OSI Model – structure

- Physical Level
- Data Link Level
- Network Level
- Transport Level
- Session Level
- Presentation Level
- Application Level

OSI Model



Physical Level: data transmission medium

Role: ensure that the sequence of bits transmitted from the transmitter reaches the receiver

– Transmission media:

- Wired (twisted pair, coaxial cable, optical fiber)
- Wireless (electromagnetic spectrum - radio, microwave, infrared, ...) -> next course



OSI Model

- **Physical Level:**

Data transmission:

- Analog (continuous values)
 - Example: telephone systems
 - Digital (discrete)
 - Example: computers

Data conversion from analog to digital and vice versa:

- Modem: digital data are transmitted in analog format
- Codec (coder/decoder): analog data are transmitted in digital format

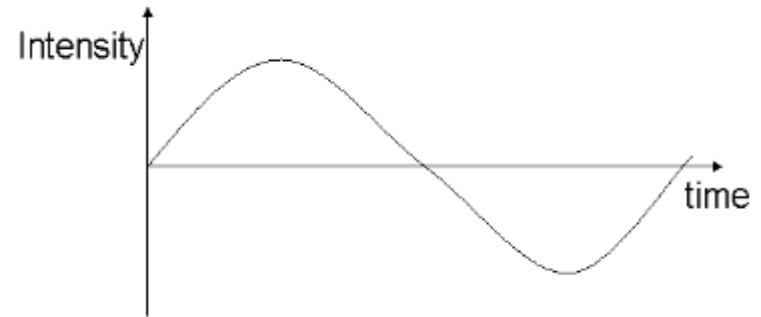


Figure. Analog Signal

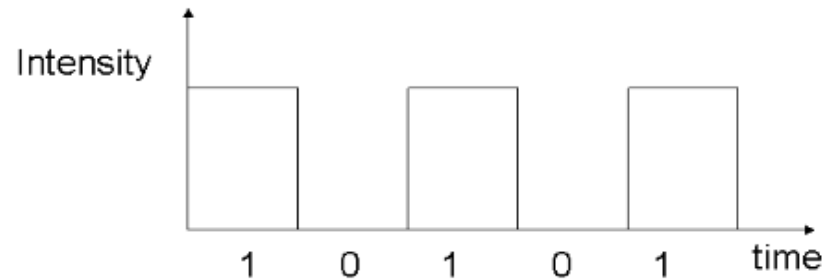


Figure. Digital Signal

OSI Model

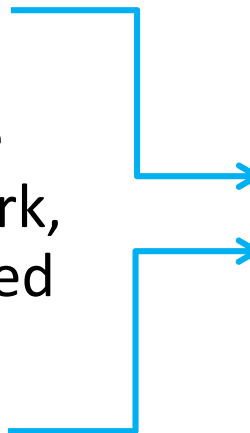
- **Physical Level - aspects**

- **Bandwidth**: the number of bits that can be transmitted over the network in a given period of time (data transfer speed)

- Usually expressed in *bits/seconds*

- **Latency**: represents the maximum time required for a bit to propagate in a network, from one end to another and it is expressed in units of time

- RTT(*Round Trip Time*) – the necessary time for a bit to cross from one end to the other and back to the environment



Basic parameters
to ensure network
performance

OSI Model

- **Physical Level - aspects**

Modification suffered by signals during propagation:

- **Attenuation:** energy loss during signal propagation through a transmission medium
- **Noise:** signal change caused by external factors (e.g. lightning, other electronic equipment, etc.)
 - Diaphony = noise from the signal transmitted by a neighbouring transmission medium
- **Distortion-** is a deterministic change of a signal



OSI Model

- **Physical Level - conclusions**

Offers transportation services, on which we can identify a number of possible problems

- Data can be altered / destroyed due to the noise
- If the destination cannot process the data in the right time, some will be lost
- If the same transmission medium is used by multiple transmitters, packages may alter each other
- It is less expensive to build logical connections to share the same physical medium, than create independent physical links



A new level?

OSI Model

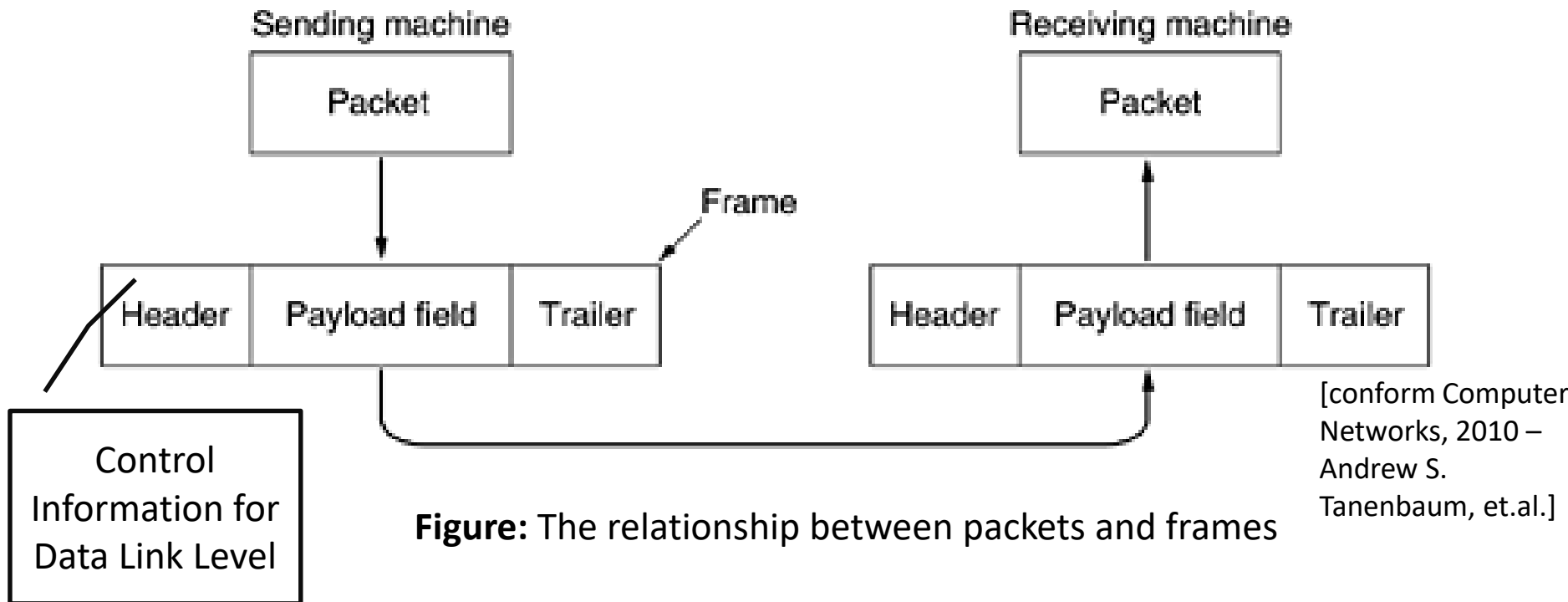
- **Data Link Level:**

- Offers:
 - mechanisms to detect and correct errors
 - regulatory mechanisms for dataflow
 - control mechanism for media access
 - Services at the network level
 - The data unit used at this level is called *frame*

OSI Model

- **Data Link Level:**

- The data are encapsulated in *frames*
- Analogy: *frame*= digital envelope



OSI Model

- **Data Link Level:**

- It provides services at the network level
 - Unconfirmed connectionless service
 - » The transmitter sends independent frames to the receiver without waiting for any confirmation
 - » A lost frame is not recovered
 - Confirmed connectionless service
 - » The sent frames are confirmed
 - » The frames are not sent in order
 - Confirmed connection-oriented services
 - » A connection is established before the transmission
 - » Frames are numbered to keep the right order

OSI Model

- **Data Link Level:**

- Divided into two sublevels:

- **LLC** (Logical Link Control)

- Role: Provides an independent view of the medium at a superior level

- **MAC** (Medium Access Control)

- Role: Used to determine who is to transmit into *multi-access channel*

OSI Model

- **Data Link Level:**

MAC (Medium Access Control)

- Context of the problem: the same physical environment is used by more emitters (uniquely identified by a physical address or MAC address) operating simultaneously, for example:
 - Half-duplex transmission between entities that use the same physical environment for both directions
 - communication by radio when there are stations that emit on the same wavelength (Wireless Ethernet - IEEE 802.11, Bluetooth, etc.)

OSI Model

- **Data Link Level:**

- MAC** (Medium Access Control)

- Strategies:

- **Static allocation**

- » FDM (Frequency Division Multiplexing)

- » TDM (Time Division Multiplexing)

- Accepting the possibility of collisions and retransmitting packets affected by collisions - **dynamic allocation**

- Collision = data is simultaneously transmitted

- General mechanism: a station that has data to send, transmit them immediately; if the collision appears, the resend action is performed

OSI Model

- **Data Link Level:**

Medium Access Control – **protocols:**

- **ALOHA**

- Pure ALOHA : *“send whenever you want”*

- **CSMA (Carrier Sense - Multiple Access):** protocol with transmission detection (*“free channel before sending?”*)

- *1-persistent CSMA*
- ...
- *p-persistent CSMA*

OSI Model

- **Data Link Level:**

- Medium Access Control – **protocols:**

- **CSMA** (Carrier Sense - Multiple Access)

- CSMA/CD (*CSMA with Collision Detection*)

- » “*free channel while transmit?*”

- » Based on Ethernet LAN (IEEE 802.3)

- **Gradually, its role diminished due to switch-based networks and full-duplex technology**

Modelul OSI

- **Medium Access Control:**

- **Hub-based networks:** When Ethernet networks are based on hubs (repeaters), CSMA/CD is essential for managing collisions that can occur when two stations try to transmit simultaneously. In these scenarios, all network stations "listen" to the same communication channel.
- **Transition to switches:** With the introduction of Ethernet switches, each connection between a station and a switch essentially became a point-to-point link => reducing the domain where collisions can occur.
- **Full-Duplex:** The modernization of networks allowed full-duplex communications, where stations can transmit and receive simultaneously, without fear of collisions => eliminating the need for CSMA/CD.
- **Gigabit and subsequent networks:** with the adoption of Gigabit Ethernet (1000BASE-T) and subsequent standards, half-duplex mode => CSMA/CD is no longer necessary.

Networks of these speeds operate exclusively in full-duplex mode.

OSI Model

- **Data Link Level:**

- Medium Access Control – **protocols:**
 - **MACA (Multiple Access with Collision Avoidance)**
 - **MACAW**
 - Improves MACA: managing medium access more efficiently
- **CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)** – Adopted in the Wi-Fi 802.11 standards (starting in the late 1990s), this is the main protocol for avoiding collisions in wireless networks, using mechanisms to prevent collisions before they occur.

The basis for wireless networks (IEEE 802.11)

Modelul OSI

- **Data Link Level:**

- Medium Access Control– **protocols -> QoS**
 - **Enhanced Distributed Channel Access (EDCA)** – Part of the IEEE 802.11e standard, introduced in the 2000s, EDCA brought prioritization for different types of traffic, providing **Quality of Service (QoS)** in Wi-Fi networks.
 - EDCA operates at the data link layer, specifically in the MAC (Media Access Control) sublayer, where it **manages traffic prioritization**. It allocates different waiting times for various types of traffic (e.g., voice, video, data), avoiding congestion and optimizing network performance.
 - **HCCA (Hybrid Coordination Function Controlled Channel Access)** – Also part of IEEE 802.11e, HCCA was introduced to implement **Quality of Service (QoS)** in wireless networks, efficiently managing channel transmission and access.
 - HCCA is a more sophisticated channel access control mechanism, allowing transmission scheduling based on stricter prioritization criteria. It is used for applications requiring very low response times, such as VoIP calls or video streaming.

Modelul OSI

- **Data Link Level** - protocole -> QoS

Enhanced Distributed Channel Access (EDCA) - is an extension of the traditional **CSMA/CA** mechanism and is designed to provide distributed channel access with priorities for different types of traffic.

Mechanism:

- **Access Category (AC):** Traffic is divided into four access categories, each with a different priority:
 - AC_VO (Voice): Very high priority, requires low latency.
 - AC_VI (Video): High priority, for video transmissions.
 - AC_BE (Best Effort): Normal priority, for general traffic (e.g., web browsing).
 - AC_BK (Background): Low priority, for traffic that is not sensitive to delays.
- **Waiting Intervals** - each access category has a different waiting interval before it can send data:
 - AC_VO has a shorter waiting time compared to AC_BE
- **Differentiated Backoff** each access category uses an adaptive backoff window (CW - Contention Window) to manage how long it waits after the channel is detected to be free.

Modelul OSI

- **Data Link Level** - protocole -> QoS
- **HCCA (Hybrid Coordination Function Controlled Channel Access)** - uses a centralized coordinator to manage channel access. This allows stricter scheduling and guarantees for QoS.

Mechanism:

- **Hybrid Coordinator (HC)** - manages access to the channel, can take control of the channel at any time, and can allocate specific transmission periods for each device.
- **Polling periods** - HC allocates polling periods where it actively requests certain devices to send data.
- **Guaranteed service intervals** - HC can ensure guaranteed service intervals for real-time voice and video traffic.
- **Efficiency for time-critical applications** - HCCA is very efficient for time-critical applications because the coordinator can schedule specific transmissions and manage channel access priorities, unlike the distributed access of EDCA.

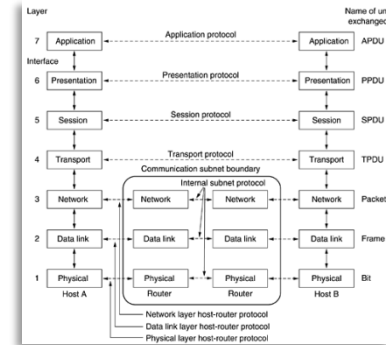
Standard IEEE	Description
802	Group standards for LAN and MAN
802.2	LLC (Logical Link Control)
802.3	Ethernet (Carrier Sense Multiple Access with Collision Detect (CSMA/CD))
802.3u	Fast Ethernet
802.3z	Gigabit Ethernet
802.11 a/b/g/n/ac	Wireless (WLAN) – with CSMA/CA
802.15	Wireless PAN (802.15.1 Bluetooth, ...)
802.16	Wireless WAN

Medium Access Control – Standards Example

OSI Model

- **Data Link Level** - equipment
 - *Bridges*
 - Resend frames between two networks (LAN)
 - It doesn't change the frame content and only headers can be modified
 - Improve safety and performance transmission
 - Can provide flow control and congestion data
 - Retransmission is done via static routes or using a spanning tree
 - STP (IEEE 802.1D) – Spanning Tree Protocol
 - **Other equipment? (Course 1)**

OSI Model



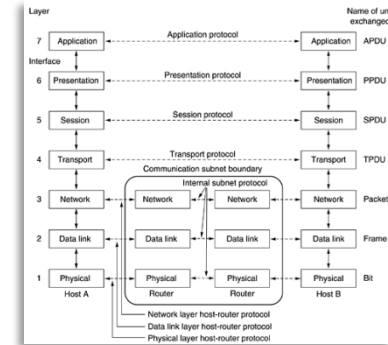
- **Network Level:**

- Retrieves packages from the source and transfer them to the destination
- It provides services to transport level
 - What services?

- Internet community proposes:

- » Connectionless services: SEND PACKET, RECEIVE PACKET
 - » Packages (called ***datagrams***) are independent and are managed individually
 - » Datagram services are similar to a typical post system

OSI Model



- **Network Level:**

- Retrieves packages from the source and transfer them to the destination
- It provides services to transport level
 - **What services?**
 - Telephone companies propose:
 - » Connection-oriented service – safe services
 - » Before the transfer some negotiations are initiated to establish a connection (VC-virtual circuit)
 - » These services are similar to the telephony system

OSI Model

- **Network Level:**

- Used Protocols

- X.25 (Connection-oriented)
 - IP

- Problems

- Protocol conversions and addresses
 - Error control (flow, congestion)
 - Dividing and recomposing packages
 - Security – encryption, *firewall*

Modelul OSI

MPLS (Multiprotocol Label Switching)

- Used to transport different types of traffic (data, voice, video) between data centers and to provide VPN services for corporate clients.
- MPLS operates between the Data Link Layer (Layer 2) and the Network Layer (Layer 3) of the OSI model,
- Mechanism
 - **Labeling** - when a data packet enters the MPLS network, a label is assigned to it by the first router (Label Edge Router - LER).
 - **Switching** - instead of using IP addresses for routing, intermediate MPLS routers (Label Switching Routers - LSRs) read the packet's label and forward it based on that, without analyzing the entire packet.
 - **Routing** - at each routing point, routers swap the label and choose the best predefined path (Label Switched Path - LSP) for the packet, ensuring fast and efficient routing.
 - **Label Removal** - when the packet reaches the exit router from the MPLS network, the label is removed, and the packet is delivered to the final destination using traditional IP routing.

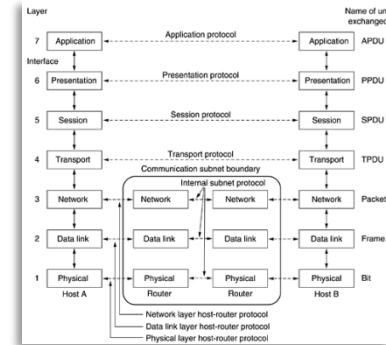
SD-WAN (Software-Defined Wide Area Networking) (nivelul 3)

- Manages traffic based on IP addresses and network-level routes, offering flexibility similar to Layer 2.5. It uses software-defined routing policies that allow dynamic selection of routes based on criteria such as latency, congestion, cost, or bandwidth.
- Does not require a dedicated infrastructure like MPLS, but uses public connections (such as the Internet or LTE) to create wide area networks (WAN) for organizations.
 - Advantages
 - Use the public Internet for traffic transport reduces costs.
 - SD-WAN can dynamically manage routes and direct traffic based on predefined policies, optimizing the use of network resources.
 - SD-WAN can combine multiple simultaneous connections (e.g., Internet and LTE), ensuring redundancy and traffic optimization to improve performance.
 - SD-WAN provides application-level visibility and can prioritize traffic based on the type of application (e.g., VoIP, video conferencing, web traffic).

Observations

- MPLS operates at Layer 2.5, using labels to switch packets more efficiently through the network.
- SD-WAN operates at Layer 3, using traditional IP routing but with intelligent routing policies and the management of multiple connections.
- MPLS requires control and a dedicated physical network provided by an operator.
- SD-WAN uses any combination of public (Internet), private (MPLS), or LTE networks to optimize traffic, reducing costs and creating a more flexible network.
- MPLS – when strict performance and guaranteed QoS are needed.
- SD-WAN – when flexibility and cost are the priority.

OSI Model



- **Transport level:** it offers safe and cost-effective data transport from the source machine to the destination machine, independent of physical network or networks currently in use

Services: provides connection-oriented and connectionless services



Differences between the transport and network layer?

OSI Model

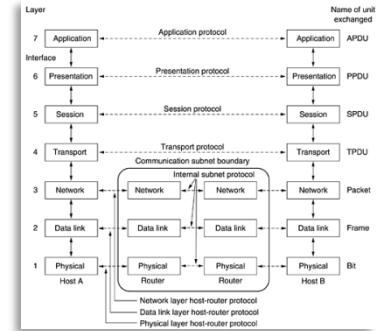
- **Transport level:**

- **Primitives:**

- LISTEN – it's a blocking operation until a process tries to connect
 - CONNECT – trying to establish a connection
 - SEND – send data
 - RECEIVE – it's a blocking operation until data is received
 - DISCONNECT – connection release

- **Performance** - quality of service (QoS - Quality of Service):
establishing /releasing the connection, error rate, protection,
priority, resilience (the probability that a connection shut down
because various internal reasons), duplicate packets, flow control

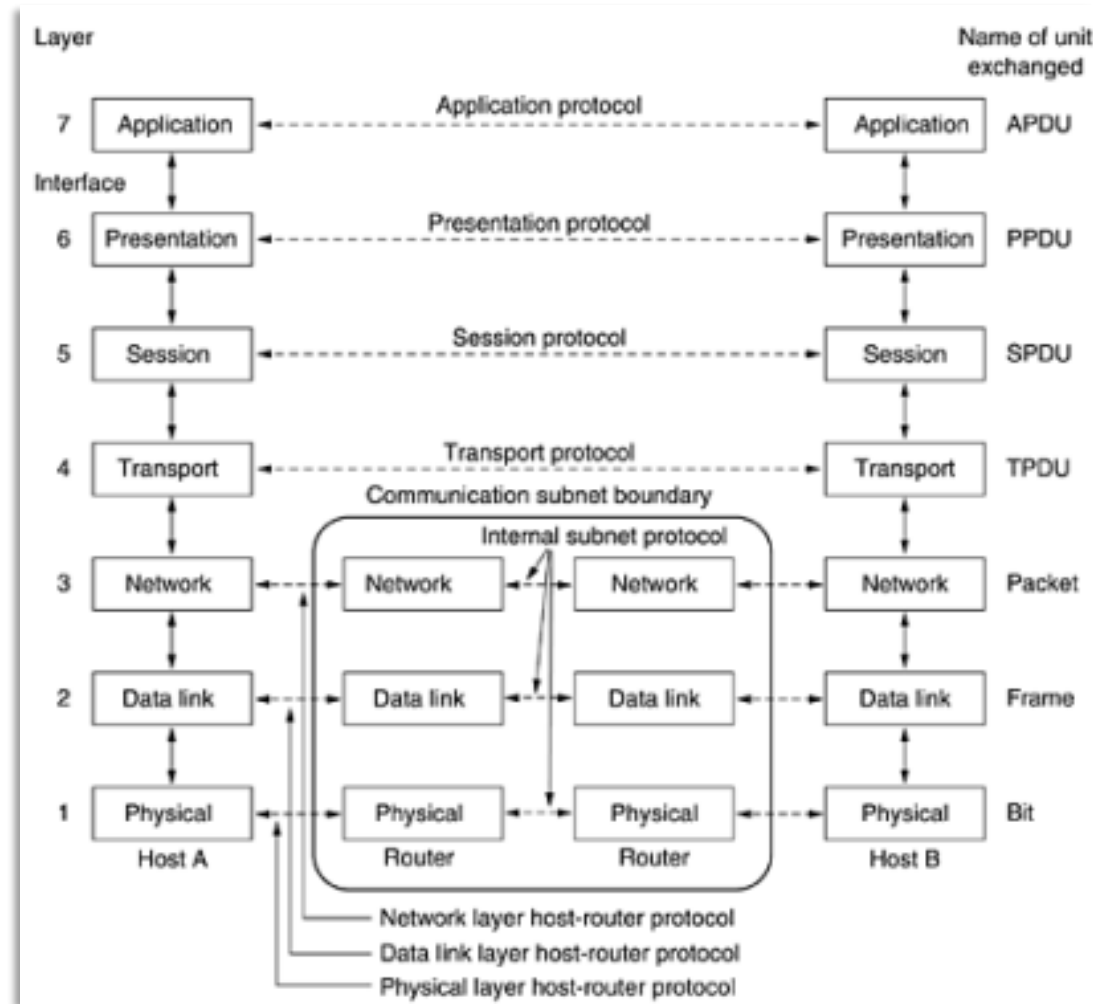
OSI Model



- **Session level:** refers to problems linked to session settings (dialogue control services, synchronisation etc.)
- **Presentation level:** handle data presentation, codified them into standard format
 - To ensure communication among computers with different representations, the presentation level ensures the conversion of internal data in standardized network representation and vice versa

Modelul OSI

- **Application level:** manage network services: virtual terminal, file transfer, electronic mail, remote execution of applications, ...



TCP/IP Model

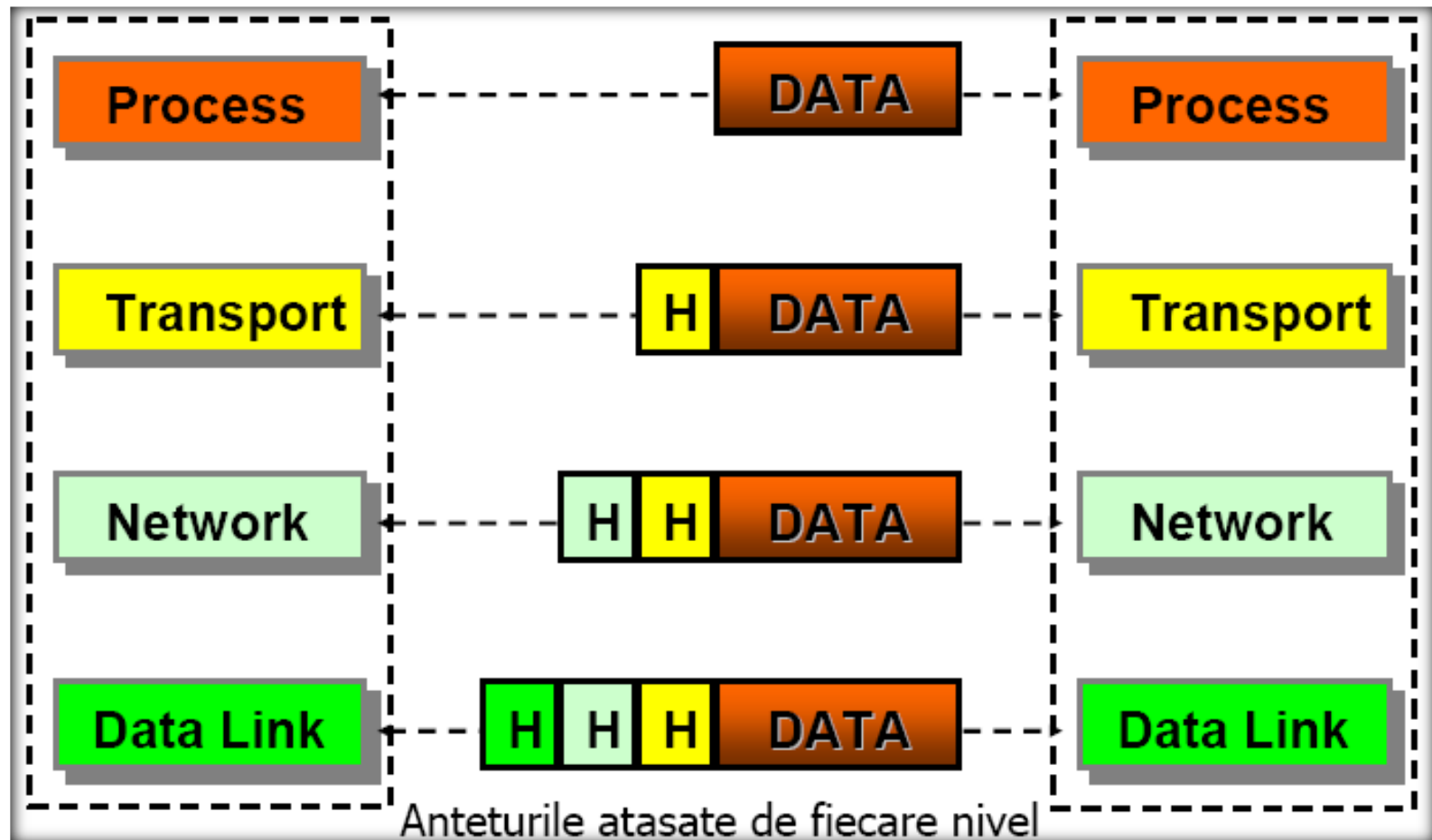
- **Terms:**
 - **end-system** – *host*
 - **network** - provides support for data transfer between end systems
 - **internet** - collection of networks (interconnected)
 - **Sub-network** - part of the internet
 - **intermediate system** - connects two sub-networks

OSI versus TCP/IP

TCP/IP Model	TCP/IP - Protocols	OSI Model
Application	FTP, Telnet, HTTP,...	Application
		Presentation
Transport	TCP, UDP, ...	Session
		Transport
Internetwork	IP, ...	Network
Host to Network	Ethernet, ...	Datalink
		Physical

Figure: Overview of models OSI and TCP / IP

TCP/IP Model



TCP/IP Model

- It provides the ability to interconnect multiple network types
- Network and Transport levels are the kernel of this model
- Successfully implemented over Ethernet (IEEE 802.3) - supported by many implementations of the physical layer (coaxial cable, twisted pair, fiber optic)

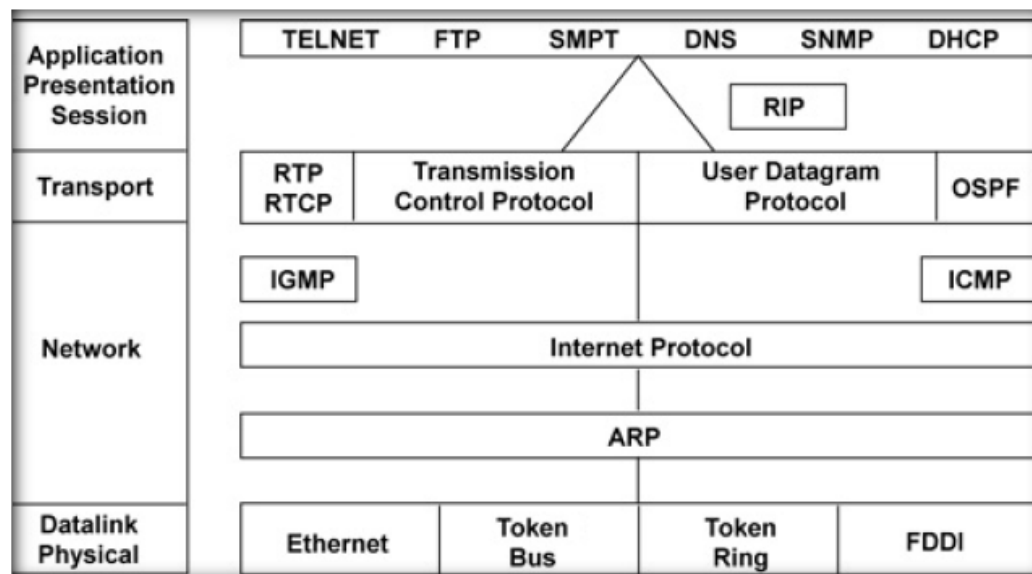


Figure. The TCP / IP - protocols

TCP/IP Model

- “Physical” level

- Ensure the connection between host and the network

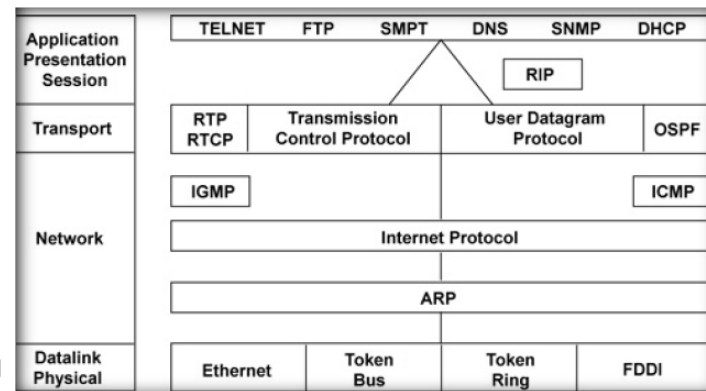
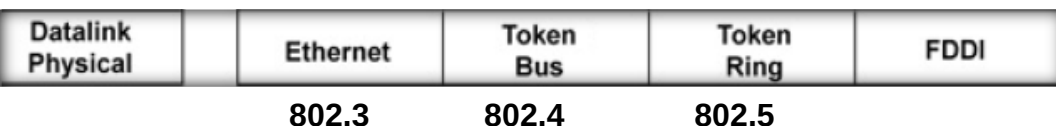


Figure. TCP/IP Model

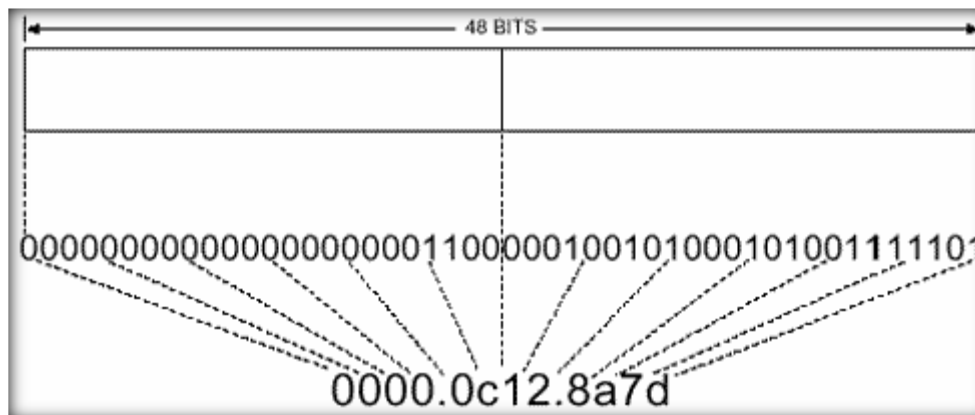
Ethernet

- It provides multiple access (shared transmission medium) in a network
- Collision Detection: CSMA / CD (Carrier Sense Multiple Access with Collision Detection)) before Gigabit Ethernet or 1000BASE-T - full-duplex
- Each Ethernet interface has a unique address 48 bits: hardware address (MAC) - e.g. C0: B3: 44: 17: 21: 17
 - Addresses are assigned to NIC (Network Interface Card) producers by a central authority

TCP/IP Model

Ethernet

- Each interface (board) network has a unique MAC address (some operating systems allow it to be modified by software)



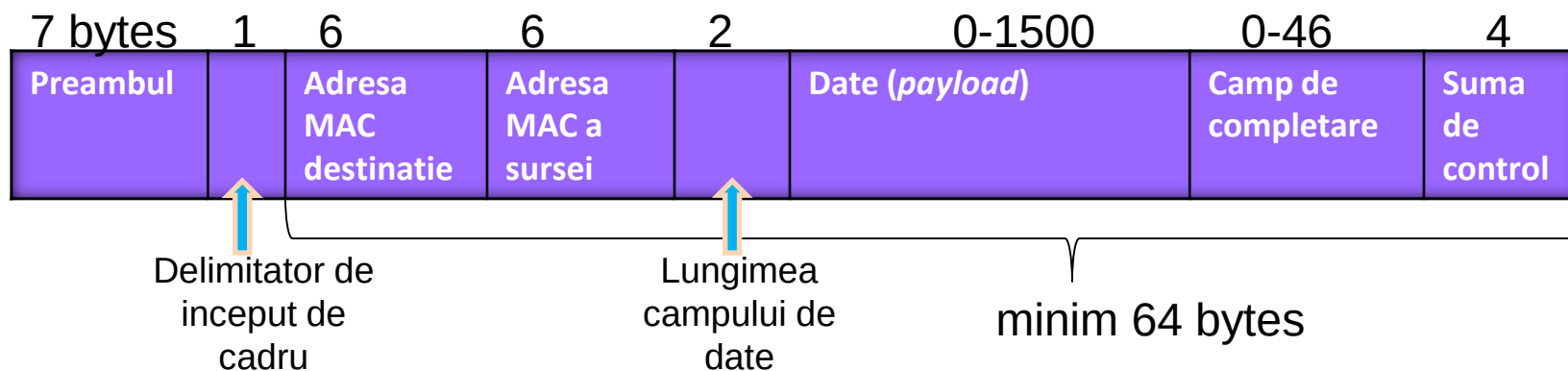
- The first 24 bits identify the manufacturer

[conform Retele de calculatoare –
curs 2007-2008, Sabin Buraga]

TCP/IP Model

Ethernet

- A frame format:



- Broadcast: the address has all bits set to 1 (FF:FF:FF:FF:FF)
- Each network interface inspecting the destination address in each frame
- If the destination address does not match with the hardware address or the broadcast address, then the frame is ignored

TCP/IP Model

Ethernet – standards (examples):

- 10 BASE5: 10 Mbps using thick coaxial cable (Thick Ethernet)- 1980
- 1BASE5: 1 Mbps using two Ethernet cables (Unshielded Twisted Pair)
- 10BASE-T: 10Mbps using 2 pairs UTP– 1990
- 10BASE-FL: 10 Mbps optical fiber with point-to-point link
- 10BASE-FB: 10Mbps backbone with optical fiber
- 100BASE – FX: 100MBps CSMA/CD with two optical fiber
- ... etc

TCP/IP Model

Ethernet versus Fast Ethernet

	Ethernet	Fast Ethernet
Viteza	10 Mbit/s	100 Mbit/s
Protocolul MAC	CSMA/CD	CSMA/CD
Diametrul rețelei	2.5 km	205 m
Topologie	Magistrala, stea	Stea
Tip cablu	Coax, UTP, fibra	UTP, fibra
Standard	802.3	802.3u
Cost	c	2*c

[conform Retele de calculatoare –
curs 2007-2008, Sabin Buraga]

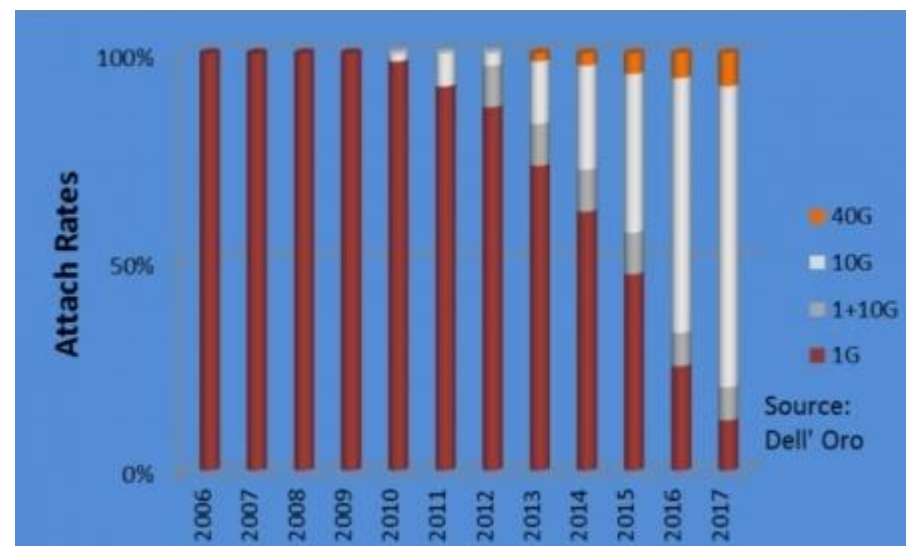
TCP/IP Model

- **Gigabit Ethernet**

- Implementations for both copper wires (802.3ab), and fiber (802.3z)
- The difference from other Ethernet implementations is at physical level

- **10 Gigabit Ethernet**

- Implementations for fiber (802.3ae)
- Operates at distances of 40km (useful for MAN and WAN)
- Frame format is similar to other implementations of Ethernet



[<http://www.networkcomputing.com/networking/will-2014-be-the--year-of-10-gigabit-ethernet/a/d-id/1234640?>]

TCP/IP Model

Standard	IEEE documentation	Cable	Minimum cable grade	Speed	Maximum distance
10GBaseSR 10GBaseLR 10GBaseER 10GBaseSW 10GBaseLW 10GBaseEW	802.3ae	MMF or SMF	N/A	10 Gbps	82 meters to 40 km
40 Gigabit Ethernet	802.3ba	MMF, SMF, and copper	N/A	40 Gbps	40 km over SMF, 7 meters over copper
100 Gigabit Ethernet	802.3bj and 802.3bm	MMF, SMF	N/A	100 Gbps	100 km

On February 11, 2021, the IEEE-SA Board approved the IEEE 802.3cu standard.^[19] On June 16, 2021, the IEEE-SA Board approved the IEEE 802.3ct standard.^[20]

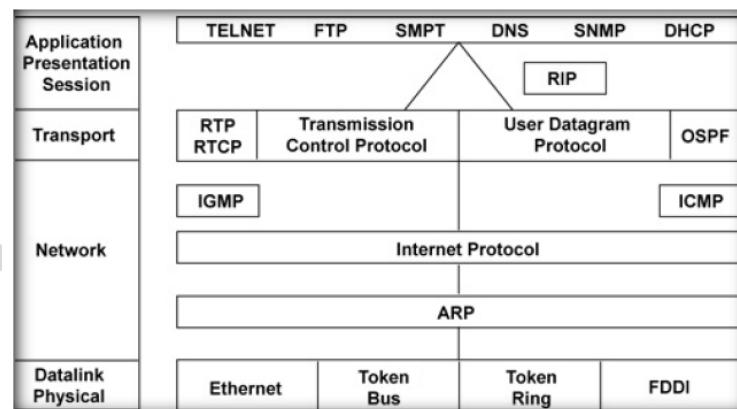
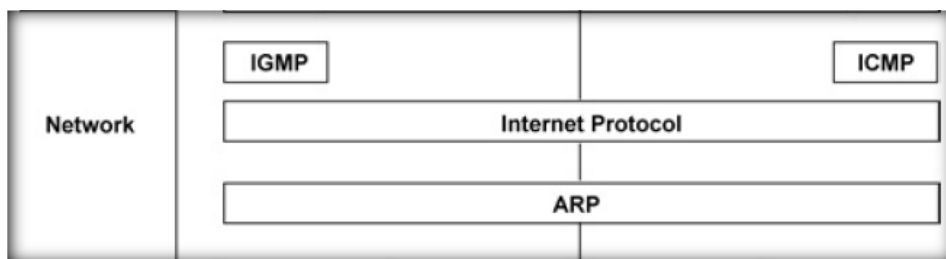
https://www.ieee802.org/3/db/P802d3db_Updated_Objectives_Approved_November_2020.pdf

[<https://www.computernetworkingnotes.com/networking-tutorials/ethernet-standards-and-protocols-explained.html>]

TCP/IP Model

- **Network Level**

- It allows hosts to emit a packet in any network; packages travel independently up to destination



- Highlights:
 - routing packets
 - congestion avoidance

TCP/IP Model

- **Network Level**
 - Level design aimed at achieving the following objectives:
 - The services offered are independent from the technology used (e.g. routers)
 - Provide transport level services, which allow it to operate independently of number, type and topology
 - It provides a unique mechanism to address LANs and WANs

TCP/IP Model

- **Network Level**

- **IPv4**
- **IPv6**
- *Routing*
 - OSPF(*Open Shortest Path First*) – RFC 1131, BGP(*Border Gateway Protocol*) – RFC 1105, EIGRP (*Enhanced Interior Gateway Routing Protocol*) - RFC 7868, RIP (*Routing Information Protocol*)-RFC 1058, RFC 2080;
- **Multicast:**
 - IGMP (*Internet Group Management Protocol*) – RFC 1112, 1054, PIM (*Protocol Independent Multicast*)- RFC 4601
- **Control:**
 - ICMP (*Internet Control Messages Protocol*) - RFC 792,777; SNMP (*Simple Network Management Protocol*) – RFC 1157; VRRP (*Virtual Router Redundancy Protocol*) – RFC 5798; RIPng (*Routing Information Protocol next generation*) – RFC 2080; ICMPv6

TCP/IP Model

- **Transport level**

- Ensures the realization of communication between the source host and destination host
- Protocols
 - **TCP** (*Transmission Control Protocol*) - RFC 793,761
 - **UDP** (*User Datagram Protocol*) – RFC 768
 - **Other Protocols: SCTP** (*Stream Control Transmission Protocol*) – RFC 4960, 3286 (2960, 3309); **DCCP** (*Datagram Congestion Control Protocol*) – RFC 4340, 4336;

TCP/IP Model

- **Application Level:**

- Contains high level protocols
- SMTP (*Simple Mail Transfer Protocol*) – RFC 5321 (821)
- POP3 (*Post Office Protocol*) – RFC 1081
- TELNET – RFC 854,764
- FTP (*File Transfer Protocol*) – RFC 454
- NFS (*Network File System*) – RFC 1095
- DNS (*Domain Name System*) – RFC 1034,1035
- HTTP (*HyperText Transfer Protocol*) – RFC 2616
- RTP (*Real-time Transport Protocol*) – RFC 3550 (1889)
- SIP (*Session Initiation Protocol*) – RFC 3261
- ...etc

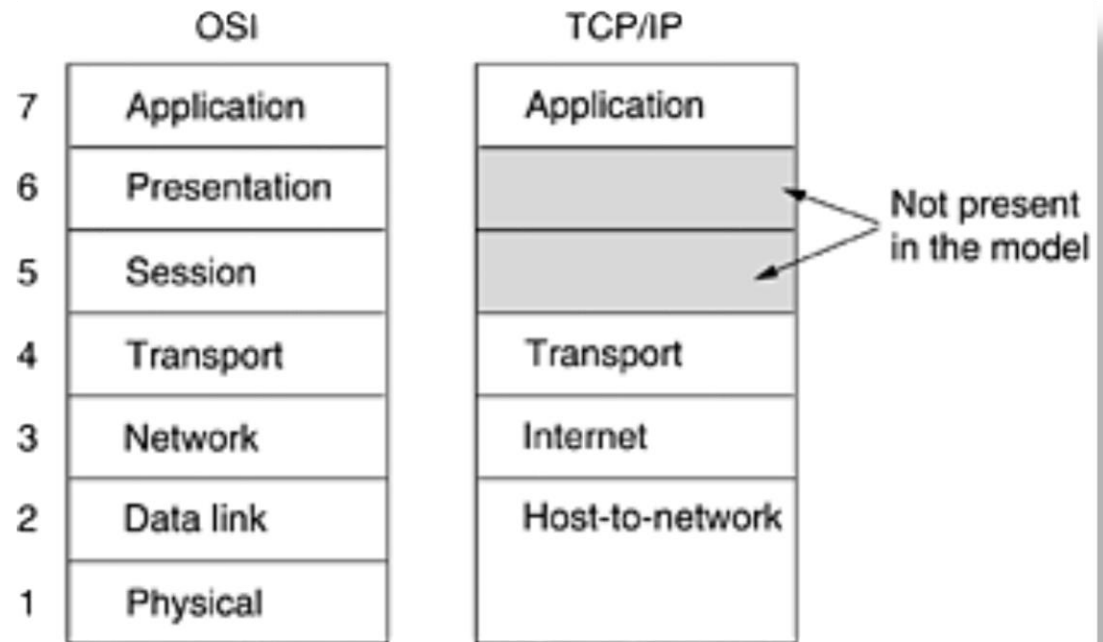
TCP/IP Model

- Organizations involved in standardization:
 - ISOC – *Internet Society*
 - IAB – *Internet Architecture Board*
 - IETF – *Internet Engineering Task Force*
 - IRTF – *Internet Research Task Force*
 - InterNIC – *Internet Network Information Center*
 - IANA – *Internet Assigned Number Authority*
- RFC (*Request For Comments*) documents
 - Edited by Network Working Group (IETF)
 - RFC 1800 (Internet Official Protocol Standards)
 - More details -> www.ietf.org

OSI versus TCP/IP

- **Similarities:**

- Both are based on a protocol stack
- The layer functionalities are somehow similar
- Both have an application layer on top
- Are based (directly or not) on transport level

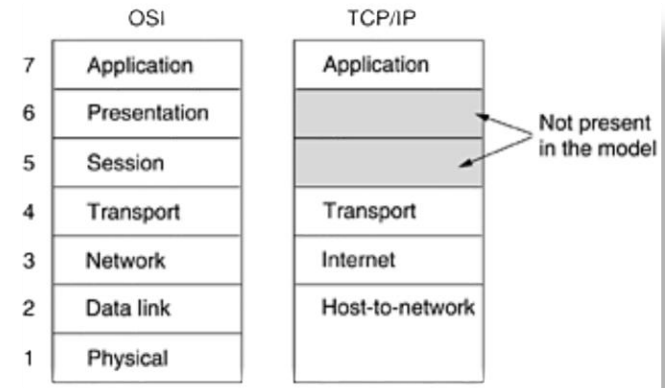


[conform Computer Networks, 2010 – Andrew S. Tanenbaum, et.al.]

OSI versus TCP/IP

- **Differences:**

- ISO/OSI is a theoretical model; TCP/IP is effective in implementation
- OSI makes explicit the distinction between service, interface and protocol; TCP / IP does not
- ISO / OSI provides protocols that ensure reliable communication (detection and treatment of errors at each level); TCP/IP verifies communication at transport level
- OSI support both types of communication at network level (connectionless and connection oriented); TCP/IP has connectionless services at network level and both types at transport level



[conform Computer Networks, 2010 – Andrew S. Tanenbaum, et.al.]

Summary

- Computer Networks Structure
- Network Architecture Models (OSI, TCP/IP)
- TCP/IP Model
- ISO/OSI versus TCP/IP

Questions?